

KUNAL PANDEY

AI Developer

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Professional Summary

AI/ML and data engineering candidate currently completing post-graduate studies in Artificial Intelligence at Seneca Polytechnic, with a completed Cloud Computing diploma from Cambrian College. Experienced building production-grade ML pipelines, RAG systems, GenAI applications, and BI dashboards using Python, LangChain, Docker, and AWS/GCP. Seeking a co-op opportunity in Ontario across data, ML, and cloud engineering domains.

TECHNICAL SKILLS

ML / AI & GenAI: Scikit-learn, PyTorch, LangChain, RAG, Pinecone, Vector Databases, Embeddings, Prompt Engineering, Hugging Face, n8n, MCP, AI Agents

Data Engineering: ETL Pipelines, Pandas, NumPy, Feature Engineering, Data Wrangling, Apify API, SQL, JSON Normalization

Cloud & DevOps: AWS (EC2, ECR, ECS, Load Balancer), GCP (Cloud Run, Vertex AI), Docker, GitHub Actions CI/CD, Container Orchestration

BI & Analytics: Power BI, DAX, Power Query, KPI Dashboards, Plotly, Streamlit, Matplotlib, Excel

Languages & APIs: Python, SQL, JavaScript, TypeScript, HTML5/CSS3, FastAPI, Flask, RESTful APIs, Gradio

Tools: Git, GitHub, VS Code, Jupyter Notebook, MLflow

PROJECTS

Domain RAG Assistant | Python, LangChain, Redis, Pinecone, Docker, AWS EC2 [GitHub](#)

- Built a multi-domain RAG assistant (Medical, ML, and LLM knowledge bases) using HuggingFace embeddings and Pinecone with metadata-based domain filtering achieving 12/12 passing retrieval evaluation tests across domain accuracy, source citation, and wrong-domain blocking.
- Implemented Redis-backed conversation memory via LangChain RunnableWithMessageHistory, enabling coherent multi-turn sessions across browser restarts with configurable TTL.
- Engineered domain-specific prompt strategies per knowledge base and built wrong-domain detection that suppresses citations for out-of-scope queries.
- Containerized the full stack (Flask + Pinecone + Redis + Gemini) with Docker and deployed to AWS EC2 via GitHub Actions CI/CD automating build, ECR push, and container relaunch on every push to main.

AI Job Recommendation System | Python, LLMs, Streamlit, Resume Parsing, APIs [GitHub](#)

- Engineered and deployed a full-stack GenAI career assistant on Google Cloud Run parsing resume PDFs with PyMuPDF, orchestrating analysis via Gemini on Vertex AI, and delivering 4 structured LLM outputs (resume snapshot, skill-gap report, 30-day roadmap, job matches) end-to-end in under 30 seconds.
- Built a real-time job fetching and recommendation pipeline integrating the Apify Global / All-Source Jobs API, computing match scores, matched skills, missing skills, and natural-language recommendation explanations across 10+ job categories.
- Implemented production-grade deployment: Docker containerization, Google Secret Manager for API credential management, GitHub Actions CI/CD for automated redeployment on every push, and Vertex AI IAM service account configuration.
- Shipped user-facing features including paginated job results, CSV export of recommendations, and a structured Streamlit interface live and publicly accessible via Google Cloud Run.

Housing Price Prediction System | Python, XGBoost, FastAPI, Docker, AWS

[GitHub](#)

- Trained an XGBoost regression model on 800K+ rows of U.S. housing time-series data, achieving R^2 of 0.87 after Optuna hyperparameter optimization across 50+ trial configurations.
- Built end-to-end ML pipelines covering data ingestion, feature engineering, preprocessing, training, and inference tracked across 10+ MLflow experiments with RMSE, MAE, and R^2 logging per run.
- Deployed via GitHub Actions CI/CD pipeline to AWS ECS/ECR with Docker and a load balancer, serving predictions 24/7 through a FastAPI REST endpoint with an interactive Streamlit visualization front-end.

PROFESSIONAL EXPERIENCE

Overnight Associate | Walmart | (April 2026– Present)

- Maintained high data accuracy in inventory tracking and stock monitoring while managing large product volumes under tight time constraints.
- Demonstrated strong attention to detail and independent work ethic, consistently meeting nightly productivity targets while pursuing full-time technical project development.
- Collaborated as part of a cohesive store team to maintain overall visual and operational store standards, adapting flexibly to changing priorities and business needs.

Overnight Associate | Metro | (Nov 2024–May 2025)

- Replenished shelves to ensure products were available and easy for customers to find.
- Applied FIFO inventory management and stock monitoring across high-SKU sections, ensuring accurate product placement and availability.
- Worked independently with minimal supervision while maintaining operational accuracy and meeting nightly throughput goals.

EDUCATION

Post Graduate Diploma – Artificial Intelligence | Seneca Polytechnic, North York, Ontario

Post Graduate Diploma – Cloud Computing | Cambrian College, Brampton, Ontario

GPA: 4.0 / 4.0

Bachelor of Technology – Electronics and Communication | Manipal University, India

GPA: 3.4 / 4.0